

Deadlock Prevention of Self-Driving Vehicles in a Network of Intersections

Florent Perronnet, Jocelyn Buisson, Alexandre Lombard^{id}, Abdeljalil Abbas-Turki, Mourad Ahmane^{id}, and Abdellah El Moudni

Abstract—Recently, new research activities have emerged for controlling traffic. Since future vehicles will travel autonomously and communicate with their surrounding environment, they will then be able to negotiate the right-of-way at intersections [cooperative intersection management (CIM)] as well as reserve their itinerary (road reservation). Both concepts, i.e., CIM and road reservation, are very promising for relieving traffic congestion. The scope of this paper is to prevent deadlock under real-time conditions, by taking advantage of these two concepts. This paper presents an appropriate graph to model the network of intersections and a sufficient condition for obtaining deadlock-free traffic with a specified route from their origin to their destination. In addition, it proposes a hierarchical approach in which a network server, intersection servers, and vehicles contribute to improving the traffic condition. Simulations were performed on a network of 25 interconnected intersections, as well as, on a real urban network to prove the effectiveness of the model. The results are presented and discussed.

Index Terms—Deadlock prevention, traffic modeling, cooperative intersection management, V2X communication.

I. INTRODUCTION

THE near future commercialization of autonomous vehicles [1] requires approaches to prevent deadlock in either an isolated intersection or in a Network of Intersections (NI). Deadlock is a situation in which no progress can be made because two or more competing actions are each waiting for the other to finish [2]. Traffic deadlock can be observed through persistent congestions that cannot be resolved without human interventions. This situation can be seen also in two conflicting left turn movements with priority to the right. The drivers perform many actions such as yield the right to pass, tighten the queue, move back, modify the itinerary or change the initial desired destination. Recovering normal traffic conditions in some cases can take many hours and even many days [3]. Instead of relying on the users of autonomous vehicles, it is worth improving the current situation in which deadlock is at the core of severe congestion.

In order to prevent the deadlock, this paper focuses on Cooperative Intersection Management (CIM). Many works

stake in wireless communication (V2X) to improve the traffic management at isolated intersections [4]. Vehicles are equipped with On Board Units (OBU) to communicate together either directly in a decentralized manner [5]–[7] or through Road Side Units (RSU) [8]–[13]. In the last case, if the RSU is coupled with an intersection manager, the CIM is called centralized. In the remainder of the paper, we consider that the traffic control at intersections is based on the centralized negotiation of the right-of-way as in [14].

CIM is different from existing traffic signalization, i.e. traffic lights, traffic signs, and priorities. First, the green is obtained through wireless communication between RSU and the surrounding OBU. Second, instead of giving the “right-of-way” to a given lane, CIM determines a sequence. More precisely, it decides which vehicle crosses the intersection first, which one is the second, and so on. Hence, the green and priorities are not displayed to all vehicles in the lane. Only vehicles that are allowed to cross the intersection simultaneously receive the green. Red and green are displayed to the driver by an onboard screen and are respected by the autonomous vehicle. Hence, CIM is able to select the exact set of vehicles that will proceed to the next intersection. This precise control of vehicles at intersections allows to rigorously consider the deadlock issue in NI.

Until a few years ago, there was no clear view on CIM operation, thus deadlock in an NI was treated through probabilistic approaches or macroscopic models [15]–[17], without using communication between vehicles and the surrounding environment. Nevertheless, deadlock results from a microscopic phenomenon. More recently many successful tests of CIM have been conducted [18]. CIM is able to avoid deadlock in a simple intersection [19] but only a few works treat the deadlock in an NI. One notable work [20] focused on the traffic deadlock through the ability of vehicles to communicate with their surrounding environment. Unfortunately, the proposed approach is limited to empty the system from current vehicles without considering new vehicles that enter the system.

This paper intends to prevent deadlock of a steady flow of vehicles without interruption. We draw the reader’s attention to the fact that deadlock is a complex problem requiring a compromise between computation time and traffic performance. Thus, the paper addresses three important issues:

1) *Computation Time*: The solution should not delay the traffic. Indeed, vehicles without a right-of-way are not allowed to cross the intersection during the computation time.

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The authors are with the Laboratoire Électronique, Informatique et Image, Univ. Bourgogne Franche-Comté, Université de Technologie de Belfort-Montbéliard, F-90010 Belfort, France (e-mail: mourad.ahmane.reserach@gmail.com).

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2) *Traffic Performance*: The traffic performance should not be heavily compromised by strong deadlock-free rules.

3) *Decentralization*: In the absolute, it would be better that the local decisions at each intersection are sufficient to converge to a deadlock-free network.

The first two issues are considered as the major requirements to provide an acceptable heuristic for preventing deadlock. In order to achieve the objective, we propose a mathematical model of the network occupancy based on the concept of reservation. Besides, this paper allows a first decentralization step through a protocol in which each intersection is able to take some decisions according to the local vehicle movements. This paper proposes a CIM protocol coupled with a heuristic able to prevent deadlock in an urban environment.

The remainder of this paper is organized as follows. Section II introduces the problem of deadlock. In section III, the mathematical modeling of the problem is described. Then, a sufficient condition to obtain a deadlock-free network is derived from the model. Section IV decentralizes the problem of traffic management and introduces the protocol for negotiating the right-of-way at each intersection. Simulation results are presented in section V. Finally, conclusions and some perspectives are given in sections VI and VII.

II. BRIEF LITERATURE OVERVIEW

Coffman *et al.* [2] have defined the four necessary conditions for a deadlock to occur:

1) *Mutual Exclusion*: A resource cannot be simultaneously shared by more than one task.

2) *Hold and Wait*: A task may request additional resources while already holding one.

3) *No Preemption*: Only the task which holds a resource can release it.

4) *Circular-Wait*: A circular chain of tasks exists where each task needs resources already held by the next task in the chain.

The authors have shown that it is sufficient to remove one of these conditions to prevent the occurrence of deadlock.

Deadlock is an important issue in many systems that involve a high degree of resource sharing between tasks. It was widely studied in different fields, such as Automated Guided Vehicle (AGV) systems in Flexible Manufacturing Systems (FMS) and parallel computing. There are four policies for handling deadlock, which are prevention, avoidance, detection and ignoring. Neither ignoring nor detection can contribute to the enhancement of the current situation of the traffic. Avoidance approaches operate before allocating the resources by checking whether or not the next states can lead to deadlock, whereas prevention eliminates a priori deadlock through an additional set of behavioral rules. For instance, if all vehicles go straight, the rule of keeping box junction free prevents the network from deadlock.

Deadlock avoidance has been widely treated in the case of some AGV that travel along built paths in FMS. The Banker Algorithm (BA) in which the resource is granted only if the next state allows all process to terminate (safe state), is the basis of several approaches. The BA was improved and extended in order to fit the transportation conditions in

FMS [21], [22]. In the general case, the detection of state safety is NP [23], [24]. Hence, the BA safety checking eliminates states that can eventually reach the termination of all jobs without deadlock. Such eliminations significantly compromise the efficiency of the management of a heavy traffic in an urban area.

Other notable works are based on Petri Net (PN) for either deadlock avoidance or deadlock prevention. In [25], a PN is used to eliminate the conflict cycle chains and accordingly routing AGV. However, there is no guarantee that the AGV will reach the destination. Hence, the approach is hard to be transposed to urban traffic management. In [26]–[28], the controlled siphons in a PN are used for deadlock prevention. The main advantage of this approach is the fact that the controlled system is deadlock-free without eliminating any safe progression of the system. Nevertheless, the minimal siphon is an NP problem. Thus it is impossible to consider such an approach for real-time application.

Doniec *et al.* [29] have noticed that in some experiments, the number of simulated deadlocks remains high compared to real traffic situations. Hence, they use the anticipation principle for which the model runs faster than the system to detect undesirable situations [30], [31]. More precisely, each driver-agent simulates the outcome of the current decision and accordingly validates it or not. However, the field of view of the driver-agent is limited to one intersection. Other simple strategies are used by traffic simulators. For instance, after a certain waiting time, cars are forced to go ahead even if the link is overloaded [32] or the priority of vehicles is modified [33]. However, deadlock is not totally eliminated in the simulation.

III. TRAFFIC DEADLOCK: CIRCULAR WAIT

The four Coffman conditions require a certain level of abstraction to fit the traffic because the vehicle movements are continuous whereas the tasks and the resources are discrete. Let us assume that a task is the occupancy of one road segment (space) in the path such that the segment dimension covers at most one vehicle (see models using cellular automata [34]). Then the first three Coffman conditions are inherent in transportation systems. Indeed, for the first two conditions, two vehicles cannot be nested and every vehicle needs an additional space for releasing the occupied segment. The third condition is subtle. Without vehicle disappearance or without the ability to move back, it is impossible to obtain a kind of preemption. Thus, once there is a circular wait, deadlock fatally occurs.

In order to highlight the deadlock problem into NI, Fig.1 presents usual situations that many drivers may encounter. Fig.1-A shows a local deadlock and NI deadlock as well, in intersections managed by traffic lights. Fig.1-B and Fig.1-C show situations resulting from either the priority to the right or traffic lights. In the case of traffic lights, the red vehicles in Fig.1-B and Fig.1-C had the green light but they were not able to move forward to free the junction when the traffic color has changed. In order to avoid such a situation, in many countries, it is forbidden to move forward in the box junction if there is not enough space on the other side.

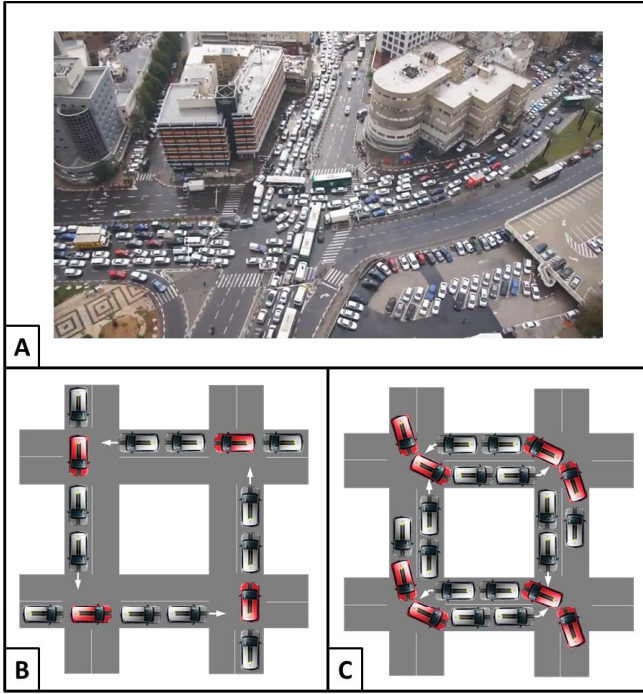


Fig. 1. Traffic deadlock situations: A- Real deadlock with intersections managed by traffic lights. B and C- Red vehicles block the four box junctions.

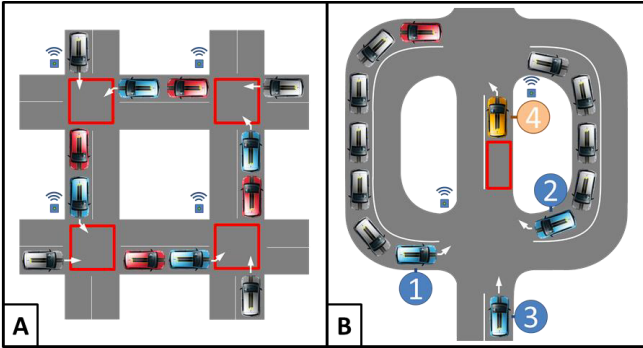


Fig. 2. A deadlock happens even if the system keeps one free place per cycle.

Let us now assume that autonomous vehicles respect the rule of box junction. This rule prevents deadlock only when all vehicles go straight. Indeed, in Fig.2-A the drivers of blue vehicles cannot move forward because the next space in the other side of the junction is occupied by red cars. Hence, each blue vehicle blocks the network by letting the box junction (red rectangle) free. Such a deadlock can be avoided if vehicles communicate together to let at least one free place in each cycle. However, in Fig.2-B, the rule is not sufficient to avoid deadlock. For instance, if vehicle 1 moves forward, one cycle is full because it will occupy the red rectangle. This happens also if vehicle 2 or 3 move forward. Vehicle 4 cannot move either because the red vehicle occupies the space on the next side of the junction. Both rules themselves block the traffic. Hence, freeing places empirically is not a proper solution to the problem.

One can observe from both figures that liveness of the network is strongly dependent on the space occupancy as well as on each vehicle itinerary. For instance, in Fig.2-B if one blue vehicle goes straight, there is no deadlock. Moreover, as shown in Fig.2-A and Fig.2-B, constraining the space occupancy is not enough to handle deadlock problem if there are turning movements. In the following, we consider that each vehicle itinerary is defined previously and the main objective of the traffic management is to control the access to the intersections to prevent deadlock. The assumption of a known itinerary is inherent to autonomous vehicles. Indeed, the vehicle should know its destination to drive by itself. Otherwise, the driver should keep his hands on the steering wheel at each intersection to guide the vehicle towards the desired direction.

A. Studied System and Problem Statement

The studied system is an NI traversed by autonomous vehicles. An NI is composed of intersections connected by roads. Intersections, as well as the roads linking intersections, have limited capacities. The roads may have one lane per direction or a bidirectional lane. The network has entrance zones and exit zones. We assume that neither entrance zones nor exit zones have limited capacities. Each autonomous vehicle has a known origin that is an entrance zone and a known destination that is an exit zone. The itinerary of each vehicle can be either determined a priori or dynamically calculated according to traffic conditions. Since the scope of the paper is to prevent the network from deadlock through CIM, we only consider the case of predefined itineraries which cannot be modified for recovering liveness. The itinerary updating process will be discussed later. Thus, the burden of the network liveness is entirely put on CIM.

In the following, we keep the anticipation principle. However, instead of considering whether or not each vehicle decision is safe, the vehicle respects the sufficient safety constraint presented hereafter. This allows avoiding the computational overhead of detecting and correcting deadlock.

The proposed model (see Fig.3) and the notations are given as follows:

- The infrastructure (Fig.3-A) is divided into n elementary spaces $\mathcal{S}_i$ (Fig.3-B), with $i = 1, \dots, n$, for which each space $\mathcal{S}_i$ cannot be physically occupied by more than one vehicle.
- We consider a three dimensional representation of the resource occupancy through space, stage (relative time step) and movement direction. Thus we have for each space $\mathcal{S}_i$, $i = 1, \dots, n$, several cells $C_{i,j,k}$ where j represents the stage (that is $j = 1, \dots, J$, with J is explained below at item d) and k denotes the movement direction (that is $\uparrow, \leftarrow, \rightarrow, \downarrow$).
- Each cell $C_{i,j,k}$ is a node of a finite graph $G = (\mathcal{C} \cup \{E_{in}, E_{out}\}, \mathcal{L})$ where $\mathcal{C}$ is the set of cells, E_{in} and E_{out} are the input and output nodes respectively and $\mathcal{L}$ is the set of directed edges (Fig.3-C).
- There is an edge from $C_{i,j,k}$ to $C_{i,j+1,k}$ if $j+1 \leq J$, with J is the number of treated stages necessary to evacuate

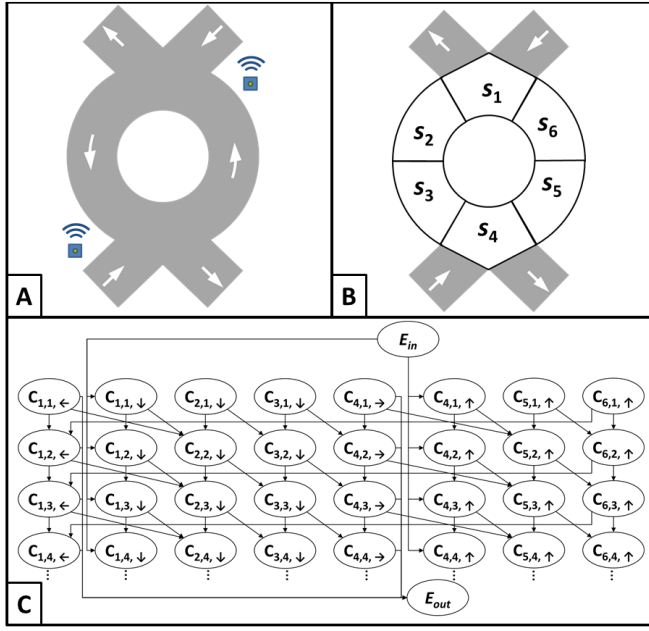


Fig. 3. Example of modeling a network of two intersections. A- The network with two intersections. B- The network is divided into six elementary spaces such that each space cannot be covered by more than one vehicle. C- The graph of cells with four stages.

the considered vehicles (For instance, in Fig.3-C we have $J = 4$).

- e) With respect to movement direction, two direct neighbor spaces S_i and $S_{i>}$ are linked through edges from $C_{i,j,k}$ to $C_{i>,j+1,k}$ under the condition $j + 1 \leq J$.
- f) We also have an edge from E_{in} to each direct reachable cell from the entrance zone as well as an edge from each cell that allows a vehicle to directly leave the NI at E_{out} .
- g) Each vehicle is designated by a number $v \in V$ where $V = \{1, 2, \dots, m\}$ denotes the set of vehicles and m is the number of vehicles. Each vehicle v is associated to a path $P(v) = (E_{in}, \bigcup_i S_i(v), E_{out})$, with $i \leq J$ and $v = 1, \dots, m$. The problem is to determine a deadlock-free path denoted $P_c(v)$ in the graph G which is given as $P_c(v) = (E_{in}, \bigcup_i C_{i,j,k}(v), E_{out})$ for each vehicle v with respect to the order of visited spaces and to movement directions (as explained above at item b).

The following subsection gives the sufficient condition of safe paths.

B. Occupancy and Reservation Constraints

Let $\mathbf{O}$ be a mapping from the set of cells $\mathcal{C}$ to the set of vehicles occupying the cells. From the definition of cells we have:

$$0 \leq |\mathbf{O}(C_{i,j,k})| \leq 1, \quad (1)$$

for all $C_{i,j,k} \in \mathcal{C}$, with $|A|$ denotes the cardinality of a set A . This means that we have either $\mathbf{O}(C_{i,j,k}) = \{v\}$ with $v \in V$ or $\mathbf{O}(C_{i,j,k}) = \emptyset$. Simply speaking, at most one vehicle can occupy one cell. By extending inequalities (1), such that:

$$0 \leq \left| \bigcup_k \mathbf{O}(C_{i,j,k}) \right| \leq 1, \quad (2)$$

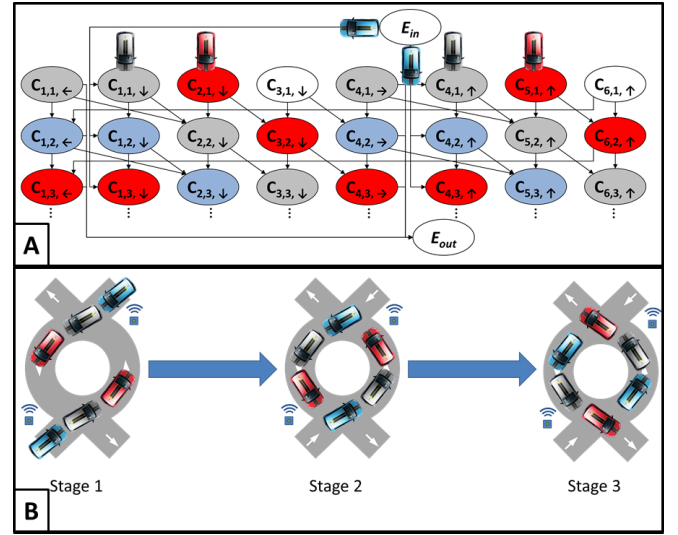


Fig. 4. Example of respected conventional occupancy constraints that lead to unrealistic movements of blue and red vehicles. A- Graph of cells with conventional occupancy constraints. B- Illustration of the three stages.

where $\bigcup_k \mathbf{O}(C_{i,j,k})$ denotes the union of all occupancy sets of cells of the same space at the same stage, then the conventional occupancy constraint is satisfied. Indeed, inequalities (2) mean that an elementary space cannot be occupied at the same stage by more than one vehicle. However, this is not sufficient to handle deadlock. Notable discussions in this modeling problem are presented in [35] and [36]. Indeed, as highlighted by the authors, occupancy constraints allow unrealistic movements as shown by the transition from stage 2 to stage 3 in Fig.4.

Instead of considering the occupancy constraint through $\mathbf{O}$, we introduce the reservation mapping $\mathbf{R}$ that is defined as follows:

$$\mathbf{R}(C_{i,j,k}) = \left(\bigcup_{k' \neq k} \mathbf{O}(C_{i,j,k'}) \right) \bigcup \left(\bigcup_{k' \neq k} \mathbf{O}(C_{i,j-1,k'}) \right). \quad (3)$$

The circular-wait happens either directly or through the transitivity of requesting the resource. The direct circular-wait occurs when two vehicles are sharing the same lane and are moving in opposite directions as shown in Fig.5-A. Formally, with $i \neq i'$ and $k \neq k'$ we have a deadlock if the three following conditions are satisfied:

$$\mathbf{O}(C_{i,j,k}) = \mathbf{O}(C_{i',j+1,k}) \neq \emptyset, \quad (4)$$

$$\mathbf{O}(C_{i',j,k'}) = \mathbf{O}(C_{i,j+1,k'}) \neq \emptyset, \quad (5)$$

$$\mathbf{O}(C_{i',j,k'}) \neq \mathbf{O}(C_{i,j,k}). \quad (6)$$

Let us now focus on the transitive circular-wait. This one is due to cycles in the NI as shown in Fig.5-B. Cycles may have several intersections in which the accessing movements are in conflict with other movements of vehicles moving in the cycle as presented in Fig.1-B. Deadlock occurs if all access spaces of the cycle are occupied and no vehicles in the cycle are leaving.

Theorem 1: If all vehicles are planned to enter then to exit the NI with respect to constraint (1) and the following

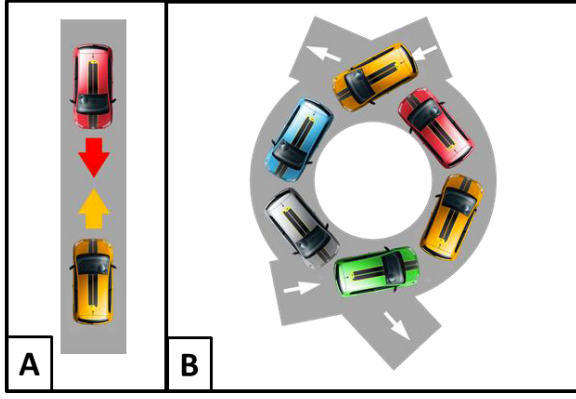


Fig. 5. Circular waits not covered by occupancy constraints. A- Direct circular wait. B-Transitive circular wait.

constraint:

$$|\mathbf{R}(\mathcal{C}_{i,j,k})| \times |\mathbf{O}(\mathcal{C}_{i,j,k})| = 0 \quad (7)$$

for all $\mathcal{C}_{i,j,k} \in \mathcal{C}$, then there is no deadlock.

Proof: We have to prove that equality (7) satisfies the occupancy constraints and prevents both circular-waits, i.e. direct and transitive as shown in Fig.5-A and Fig.5-B.

First, if equality (7) is satisfied then the occupancy constraint is satisfied too. Indeed, it is impossible to have $\mathbf{O}(\mathcal{C}_{i,j,k}) = \{v\}$ and $\mathbf{O}(\mathcal{C}_{i,j,k'}) = \{v'\}$ because if such a case exists then $\mathbf{R}(\mathcal{C}_{i,j,k}) = \{v'\}$ and $\mathbf{R}(\mathcal{C}_{i,j,k'}) = \{v\}$ which contradicts equality (7). In other words, two vehicles from conflicting movements cannot occupy the same space at the same time.

Now let us consider that a **direct circular wait** is possible, then from condition (1) and equality (3) we obtain $\mathbf{R}(\mathcal{C}_{i,j+1,k'}) \neq \emptyset$. However, from equality (5) we have $\mathbf{O}(\mathcal{C}_{i,j+1,k'}) \neq \emptyset$. Then, $|\mathbf{R}(\mathcal{C}_{i,j+1,k'})| \times |\mathbf{O}(\mathcal{C}_{i,j+1,k'})| \neq 0$. Equality (7) prevents direct circular wait. This leaves only the case of the transitive circular wait.

Let us consider that a **transitive circular wait** is possible. We assume that there is a saturated cycle, in which a vehicle v is the last vehicle entering the cycle. The vehicle v is in an access cell $\mathcal{C}_{i,j,k}$, otherwise there is no deadlock. For the same reason, behind v , there is a vehicle v' in the cycle. v' occupies $\mathcal{C}_{i',j',k'}$. There are two scenarios. In the first scenario, v' can exit the cycle from $\mathcal{S}_{i'}$. In such a scenario, there is no deadlock. In the other scenario, v' would reach $\mathcal{S}_i$. This is impossible because $\mathbf{R}(\mathcal{C}_{i,j+1,k}) = \{v\}$. ■

Theorem 1 means that each reserved cell according to equation (3) cannot be occupied. However, cells can be reserved by several vehicles. Theorem 1 is powerful because it identifies safe states according to whether or not equality (7) is satisfied. In the example given in Fig.4, the reservation constraint is not respected. There are many feasible solutions that respect equality (7). One solution is given in Fig.6. According to Theorem 1, the blue vehicle at the left side cannot have an access to space $\mathcal{S}_4$ before the three vehicles coming from the other side leave the space. Indeed, for occupying $\mathcal{C}_{4,2,\uparrow}$, the blue vehicle must be able to reserve both cells $\mathcal{C}_{4,2,\rightarrow}$ and $\mathcal{C}_{4,3,\rightarrow}$. However, the latter ($\mathcal{C}_{4,3,\rightarrow}$) is occupied

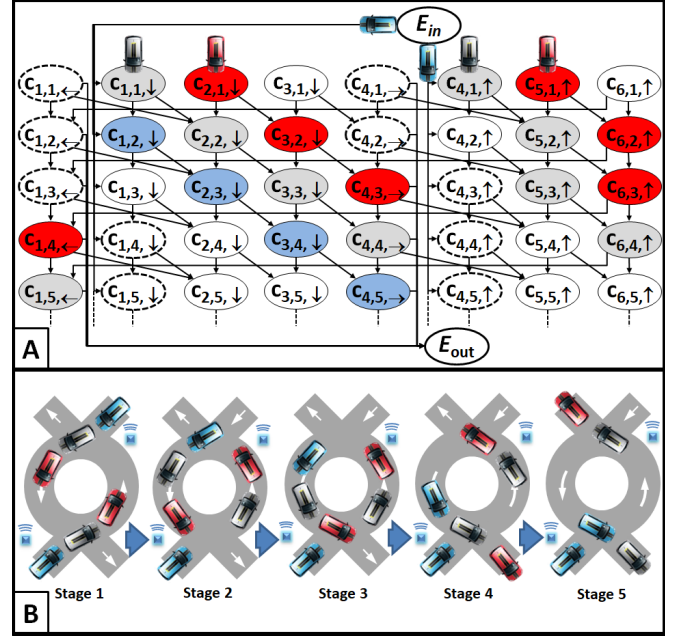


Fig. 6. Example of the reservation constraints. A- The graph with reserved cells (dotted) that cannot be occupied. B- Illustration of the five stages.

by the red vehicle. It cannot neither occupy cells $\mathcal{C}_{4,j,\uparrow}$, with $j = 2, \dots, 6$ because they are reserved by the three vehicles that occupy cells $\mathcal{C}_{4,j',\leftarrow}$, with $j' = 2, \dots, 5$.

C. Proposed Approach

Instead of checking equality (7) at each new progress of the system, a safe path $P_c(v)$ in G is provided to each vehicle. Hence, at each new incoming vehicle, the network server determines a safe path for the vehicle, which consists of finding $P_c(v)$ in G :

- By ignoring all already occupied and reserved cells in G for occupation,
- Without reserving already occupied cells.

The problem of planning one vehicle is polynomial because it has at most the complexity of the shortest path problem in a graph without negative weights. Hence, the complexity of planning one vehicle is upper bounded by $\mathcal{O}((|\mathcal{C}| + 2) \times |\mathcal{L}|)$ where $|\mathcal{C}|$ is the number of cells and $|\mathcal{L}|$ is the number of arcs. However, this requires a server for the global network. The server must provide the computation of path and the traffic signalization to vehicles. The following section shows how the global problem is decomposed in order to lighten the duty of the global server.

IV. TRAFFIC CONTROL

The traffic control is based on CIM in which vehicles negotiate the right-of-way through the intersection server as shown in Fig.7. Many demonstrations of CIM have been conducted for a simple intersection [37]. However, a network server is required to prevent the NI from deadlock. In the following, we consider a network server that communicates with each intersection server as shown in Fig.8 and Fig.9.

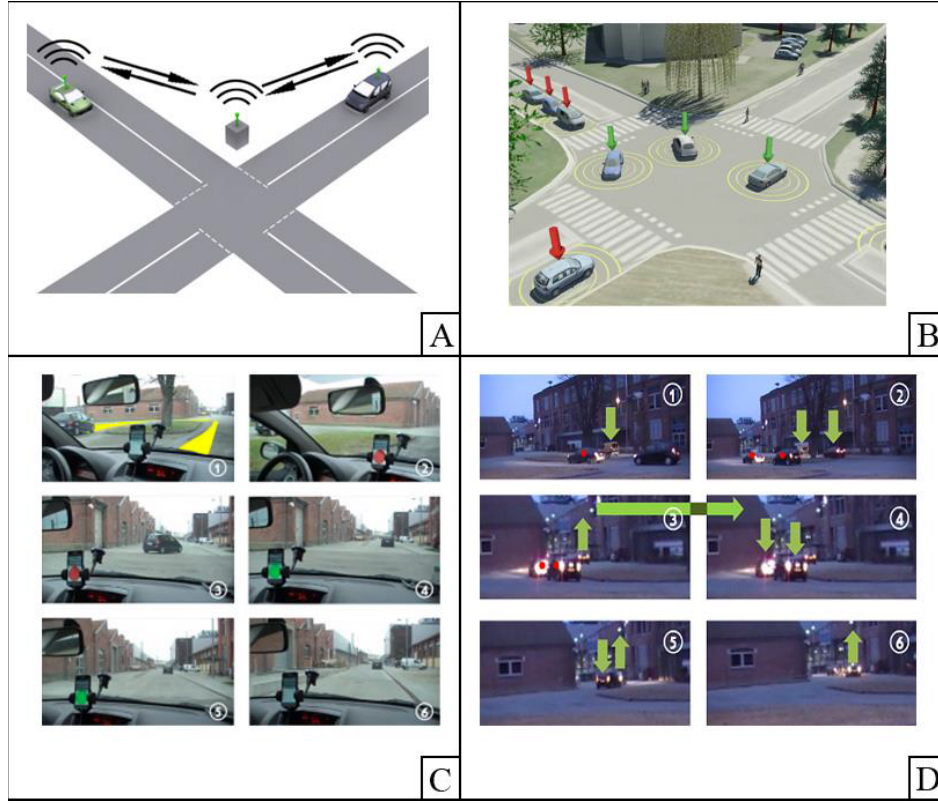


Fig. 7. Cooperative intersection management. A- Two vehicles negotiate the right-of-way through the intersection server. B- Simulation of cooperative intersection management (Voxelia Simulate). C- Real tests of negotiation between two vehicles through a server (Onboard view). D- Outside view of real tests of cooperative intersection management.

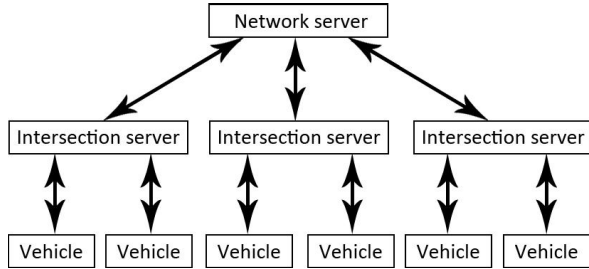


Fig. 8. Architecture of the deadlock prevention protocol: There are two ranks of controllers which are network server and intersection servers.

A. Decomposition Strategy

Both levels of traffic control, i.e. network and intersection servers, aim at lightening the duty of the whole management of the traffic. Instead of considering the entire graph G of NI as presented previously, the monitored NI is decomposed into four different zones that are entrance, conflict, storage and exit zones. Conflict zones designate the intersections whereas storage zones designate the lanes that link the considered intersections. Entrance and exit zones are lanes that allow vehicles to enter the system or to exit. More detailed decomposition rules are as follows:

- 1) The vehicle itinerary is considered as an upstream problem treated separately. Hence, each vehicle knows its itinerary that is a set of ordered spaces.
- 2) The network server considers only storage zones (lanes) between the intersections (conflict zones).

Hence, only spaces of storage zones are modeled to check occupancy and reservation constraints whereas the accesses to conflict zones are treated by the intersection servers.

- 3) In each lane (storage zone) between two intersections, one space is virtually removed. This space can be physically occupied by vehicles, but it is ignored by the network server. This allows considering the reservation constraint and avoiding a transitive deadlock situation.
- 4) The network server ignores the entrance and exit zones. However, it keeps communication with the entrance intersection servers to preserve the order of vehicles and with the exit intersection servers to remove vehicles.
- 5) Intersection servers respect the order of vehicles that have the same next storage zone as current destination.
- 6) A follower vehicle is added by the network server to the same stage or to a higher stage of its leader.

The first rule allows the network server considering only two directed edges per cell $(C_{i,j,k})$ for finding each vehicle's path $P_c(v)$. Indeed, the vehicle v that occupies $(C_{i,j,k})$ will:

- either stay in the same space S_i by occupying $C_{i,j+1,k}$,
- or move to the unique next space $S_{i'}$ of its itinerary by occupying $C_{i',j+1,k}$.

Moreover, both second and third rules considerably reduce the initial graph G by removing several cells, including those which model the various movement directions at intersections (conflict zones).

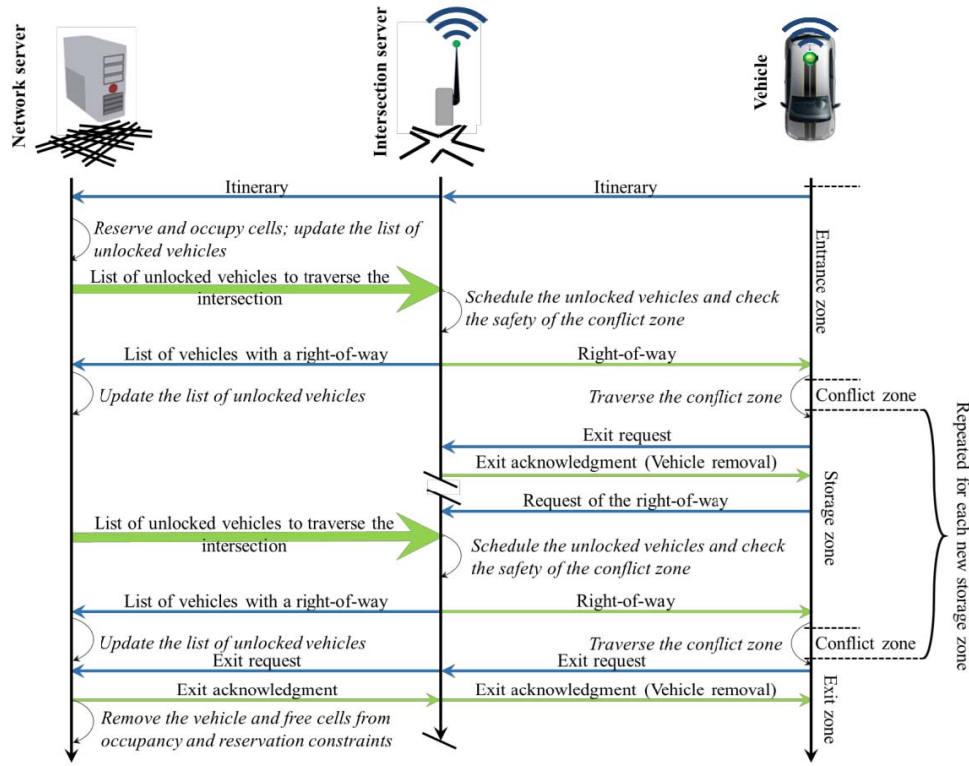


Fig. 9. Messages during the journey of a vehicle for traversing the network.

B. Architecture of the Deadlock Prevention Protocol

For safety and efficiency reasons, each intersection is equipped with a server that communicates with the vehicles, in order to distribute the rights-of-way. The intersection server selects which vehicles have the right-of-way and sends the green to them. In the proposed architecture we consider a network server which communicates with all intersection servers to allow vehicles to continue their journey. However, the control of conflict zones is the duty of intersection servers. The architecture of the protocol is depicted in Fig.8 and detailed as follows:

1) *Network Server*: It collects data on each vehicle itinerary and updates the data according to the current and future movements. The server builds suitable data structures that store paths according to occupancy and reservation constraints. The network regularly gives to the intersections the list of vehicles which are no longer under constraints, according to presence lists. These vehicles are called unlocked vehicles. No right-of-way can be given by an intersection server without a prior consent of the network server.

2) *Intersection Server*: It collects information from vehicles that envisage crossing the intersection and sends it to the network server which determines the unlocked vehicles. Then, using the list of unlocked vehicles, the intersection decides which vehicle will receive the right-of-way. Thus, the intersection server keeps a certain amount of freedom to improve the local evacuation time.

3) *Interactions*: The network server communicates with intersections and the intersection servers communicate with vehicles. The journey of a vehicle can be described through the following four subparts (Fig 9):

- **Entrance**: Once the vehicle detects the first entrance zone of the network, it sends an enter request which describes the desired itinerary to the intersection server. The request is then transmitted to the network server. The network server adds the vehicle in its database according to occupancy and reservation constraints,
- **Getting the right-of-way**: Once the vehicle is unlocked by the network server, the current intersection server is informed. According to local safety and efficiency criteria, the intersection server sends the right-of-way to the corresponding unlocked vehicle. At the same time, it notifies the network server in order to update the list of unlocked vehicles,
- **Access to a new storage zone**: Once a vehicle actually leaves the intersection, it sends an exit request and receives an acknowledgment. Then, the vehicle sends a request of the right-of-way to the next intersection server,
- **Exit**: Once the vehicle reaches one exit zone, it sends a final exit request to the network server through the intersection server. The vehicle is completely removed from the network server.

The next subsection gives more details about updating and unlocking vehicles.

C. Reservation and Occupancy Constraints Through the Protocol

According to the network server availability, at each new arrival of a vehicle v , the network server computes the path $P_c(v)$ that respects the occupancy and reservation

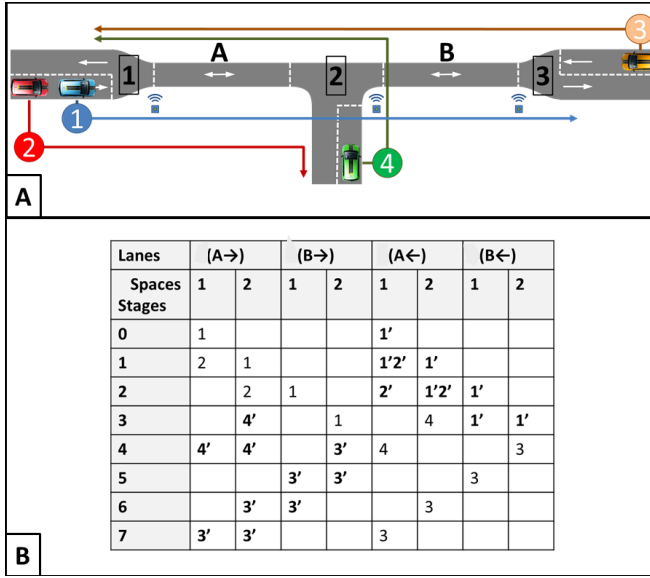


Fig. 10. Example of application of occupancy and reservation constraints. A- Four vehicles attempt to cross the network with different itineraries indicated by arrows; they arrive successively 1, 2, 3 and 4 in a short amount of time. B- Occupancy and reservation constraints with v and v' in each entry indicating that vehicle v either occupies or reserves the cell, respectively.

constraints. As previously stated, the treated cells are limited to the lanes connecting the intersections. One space is removed per lane. In the example given in Fig.10, only lanes A and B are considered by the network server. Each lane is split into 3 spaces but the server considers only 2 spaces for each lane. Lanes A and B are bidirectional, then the server considers for each lane four different cells at each stage. One can observe in Fig.10-B that there is no cells occupied by two vehicles and that each marked cell is either occupied or reserved. For instance, according to equation (3), vehicle 1 reserves the first space of $A←$ at both stages 0 and 1 because it occupies the space $A→$ at stage 0. As stated by Theorem 1, the cells reserved by vehicle 1 can be reserved by vehicle 2 but cannot be occupied. From Fig.10-B one can observe that the proposed solution of the server is that vehicles 1 and 2 cross intersection 2 first before freeing vehicle 4. Vehicle 3 crosses lane B only after vehicle 1 crosses intersection 3.

The three required steps for allowing a vehicle to traverse the NI are detailed as follows:

1) *Updates*: The network server updates the occupancy and reservation of cells when vehicles obtain the right-of-way in the lanes controlled by the network server. The update removes the vehicles from the cells belonging to the lane in which they have obtained the right-of-way to cross the intersection. For instance, in Fig.10, the server updates the constraints when a given vehicle obtains the right-of-way in zones A and B. The occupancy and reservation of cells presented in Fig.10-B are not modified until vehicle 1 in zone A obtains the right-of-way to cross intersection 2. Fig.11-B gives the updated path when vehicles 1 and 2 have obtained the right-of-way to cross intersection 2. One can observe that both vehicles are removed from zone A.

2) *Unlocked Vehicles*: A locked vehicle is not considered by the intersection server and thus, it is not allowed to obtain

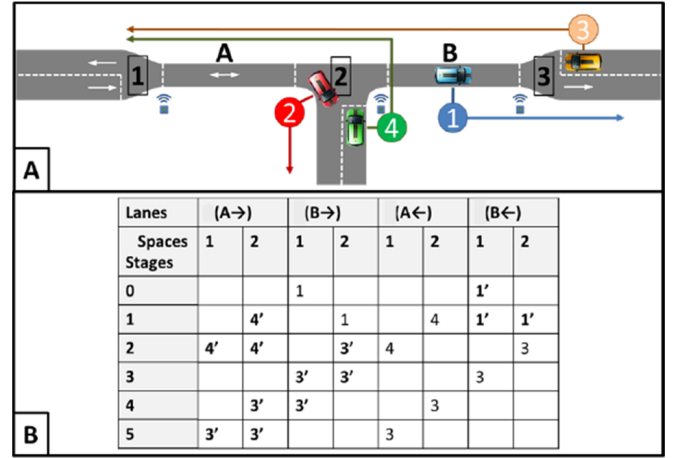


Fig. 11. Example of updated occupancy and reservation constraints. A- Vehicle 1 is in zone B requesting for the right-of-way to traverse intersection 3, whereas vehicle 2 is in zone A and has already obtained the right-of-way to traverse intersection 2. B- Updated occupancy and reservation constraints.

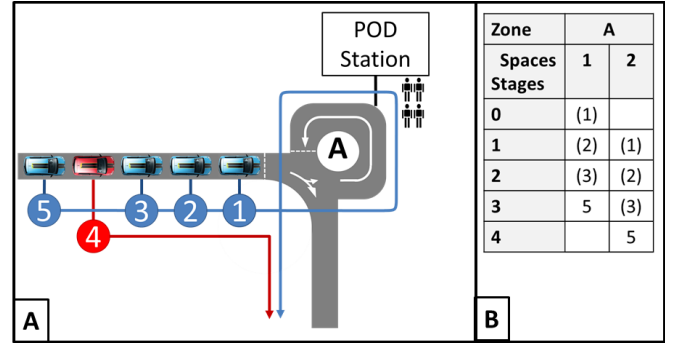


Fig. 12. Example of unlocked vehicles. A- Five vehicles attempt to cross the elementary intersection network with different itineraries indicated by arrows. B- Occupancy constraints with v and (v) in each entry indicating that vehicle v is either locked or unlocked, respectively.

the right-of-way. The vehicle in the last lane of its journey is unlocked by default. For instance, if we consider the example presented in Fig.11, when vehicles 2, 3 or 4 are in lane A, they are unlocked by default. The same rule applies to vehicle 1 in lane B. In all other cases, vehicles are unlocked only if all the following conditions are satisfied:

- All precedent vehicles occupying the zone are unlocked,
- All precedent cells are not reserved,
- The number of unlocked vehicles does not exceed the capacity of the target lane.

Let us consider the example presented in Fig.12.

In this example, lane A has one direction and is able to admit three stopped vehicles. One can note that vehicle 4 is not in the list, it is unlocked by default. Whereas, vehicles 1, 2 and 3 are unlocked because the three conditions are satisfied. Vehicle 5 is locked since the capacity of lane A is 3.

3) *Right-of-Way*: The intersection server not only prevents the vehicles from colliding but it also manages the access to neighbor storage zones. Hence, a vehicle is able to obtain

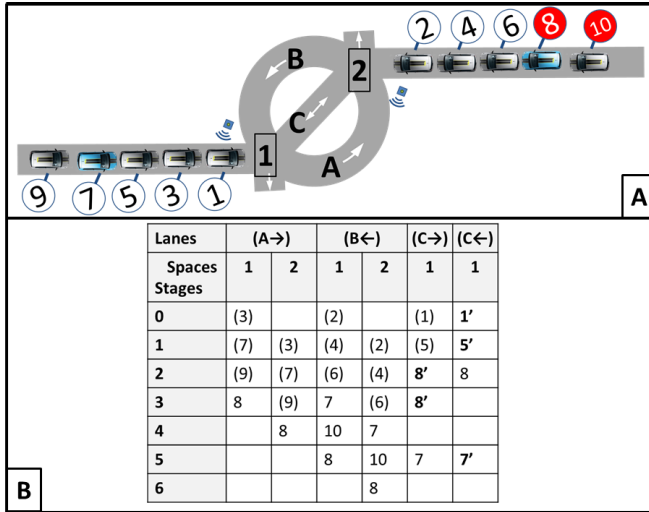


Fig. 13. Example of a network of two intersections and three lanes. A- Three connected lanes (A, B and C) throw two intersections (1 and 2): lane C admits bidirectional traffic. The color of the circle gives whether the vehicle is locked: if white the vehicle is unlocked, if red the vehicle is locked. B- The itineraries of vehicles and their planning according to the network server.

the right-of-way only if the three following conditions are satisfied:

- There is no collision risk,
- The access to the next zone is unlocked,
- The order of vehicles in the next lane is respected.

For instance, in Fig.12, the intersection server has a wide range of choices between evacuating by convoy or one by one. The updating and unlocking rules limit the convoy to at most three vehicles on lane A. Indeed, if the intersection server accepts more than three vehicles, then a deadlock will happen. More details about the wide range of choices of the intersection server are discussed hereafter through the example given in Fig.13.

D. Example

Let us consider the NI presented in Fig.13. The vehicles are numbered in the order of their arrivals. Except for both blue vehicles 7 and 8 that have to make a tour and visit the three lanes (police cars, city cleaners, touristic vehicles...), all other vehicles cross each intersection once by taking one lane, i.e. A or C for odd-numbered vehicles 1, 3, 5 and 9 and B or C for even-numbered vehicles 2, 4, 6 and 10. The itinerary of each vehicle is shown in Fig.13-B.

One can observe from this Fig.13 that all odd-numbered vehicles are unlocked while only the three first even-numbered vehicles are unlocked. Indeed, vehicle 8 is unlocked only after vehicle 5 leaves C and frees the reserved cells. This rule avoid deadlock in C.

Since initially there is no conflict in intersections 1 and 2, let us consider that all unlocked vehicles have received the right-of-way. Fig.14 shows the result.

When the vehicles in lane B receive the right-of-way, one can observe that the server of intersection 2 has a wide choice of sequences formed by all odd-numbered vehicles to choose from. If all vehicles of lane C get the right-of-way first,

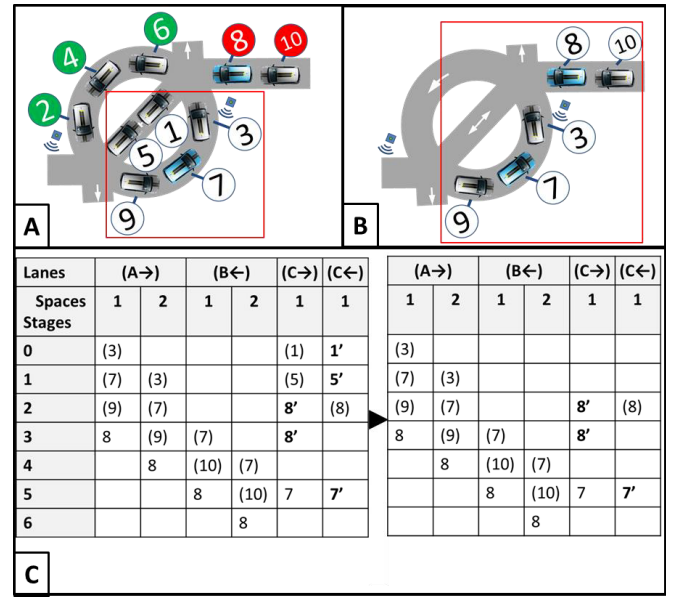


Fig. 14. The two stages after vehicles 1, 3, 5, 7 get the right-of-way from intersection 1 and vehicles 2, 4 and 6 get the right-of-way from intersection 2. A- The first stage. B- The second stage after vehicles 2, 4, 6 get the right-of-way from intersection 1. C- The itineraries of vehicles and their planning according to the network server.

the server of intersection 2 can choose between all present vehicles under the condition that vehicle 7 must pass through the intersection before vehicle 10, as shown in Fig.15-C.

E. Improvements

The proposed approach is an heuristic derived from Theorem 1. The heuristic allows lightening the duty of the network server. The proposed algorithm can be improved in order to enhance the traffic throughput or to further minimize the computational strain on the server by reducing the number of cells.

1) *Traffic Efficiency*: The update operation can be extended to compute new paths of vehicles according to their new positions. If we consider the example presented in Fig.13 and Fig.14, when vehicles 1 and 5 get the right-of-way, the NI server can update the paths of vehicles 8 and 10 as shown in Fig.15-A. Fig.15-B and Fig.15-C compare the results with and without an update of vehicle paths, respectively. With updated paths, vehicle 10 is able to leave the network without waiting for vehicle 7 as shown in Fig.15-B. Indeed, the vehicle 7 is locked in lane B (i.e. it cannot get the next destination), until vehicle 8 leaves lane C. The update can also be extended to cover the case of dynamically computed itineraries as discussed in [38].

2) *Number of Cells*: The number of columns per lane and per direction can be reduced to one column without any deadlock risk. When the size of the network is small and there is not an important demand, this will not change the result, as in all previously treated examples. However, there exist many other traffic situations for which small number of columns reduce the number of vehicles in the network. The example presented in Fig.16 shows the impact of the considered number of columns.

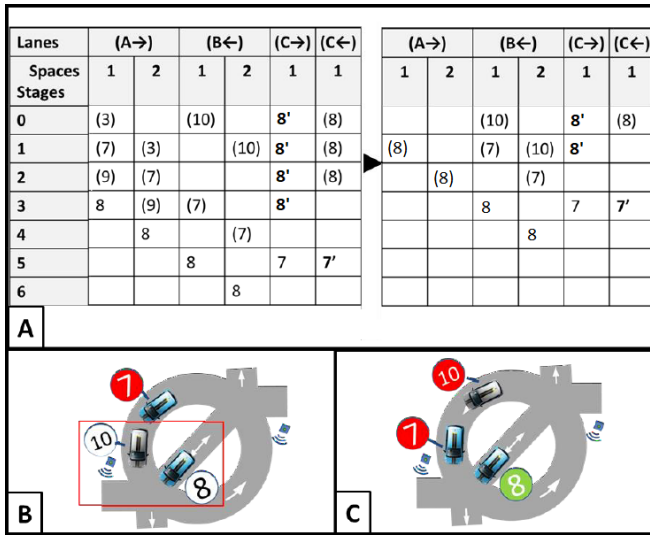


Fig. 15. Comparison between whether or not the network server computes new paths, A- Updated paths, B- The stage with updated paths, C- The stage without updated paths.

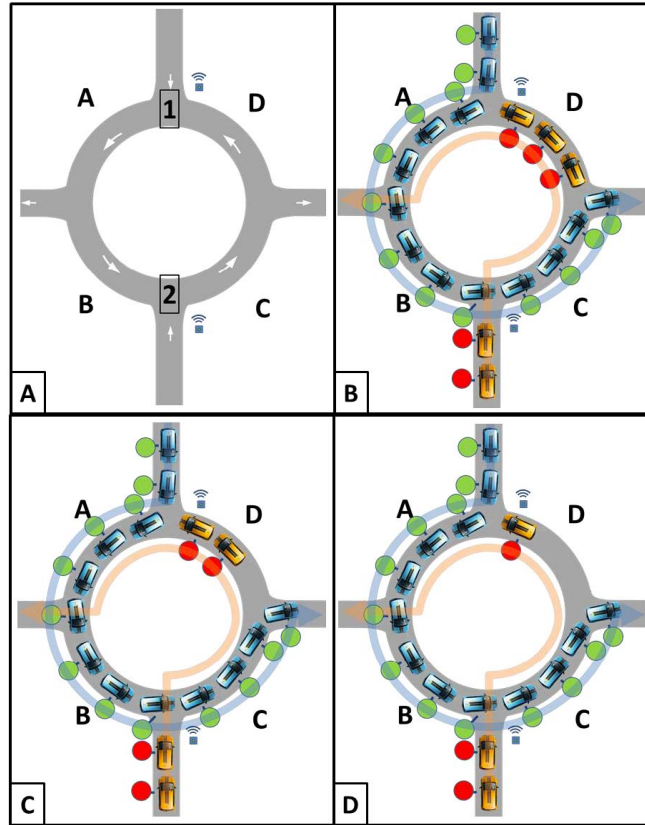


Fig. 16. Comparison between different numbers of considered columns, A- Example of NI. The numbers of admitted yellow vehicles according to: B- The theorem, C- The heuristic and D- One column per lane.

More precisely, the number of columns determines the number of last planned vehicles in the network. Hence, there is a tradeoff between traffic efficiency and the size of the graph i.e., the overall computational time.

There are various ways to optimize the traffic by updating the vehicle paths. For instance, the algorithm allows

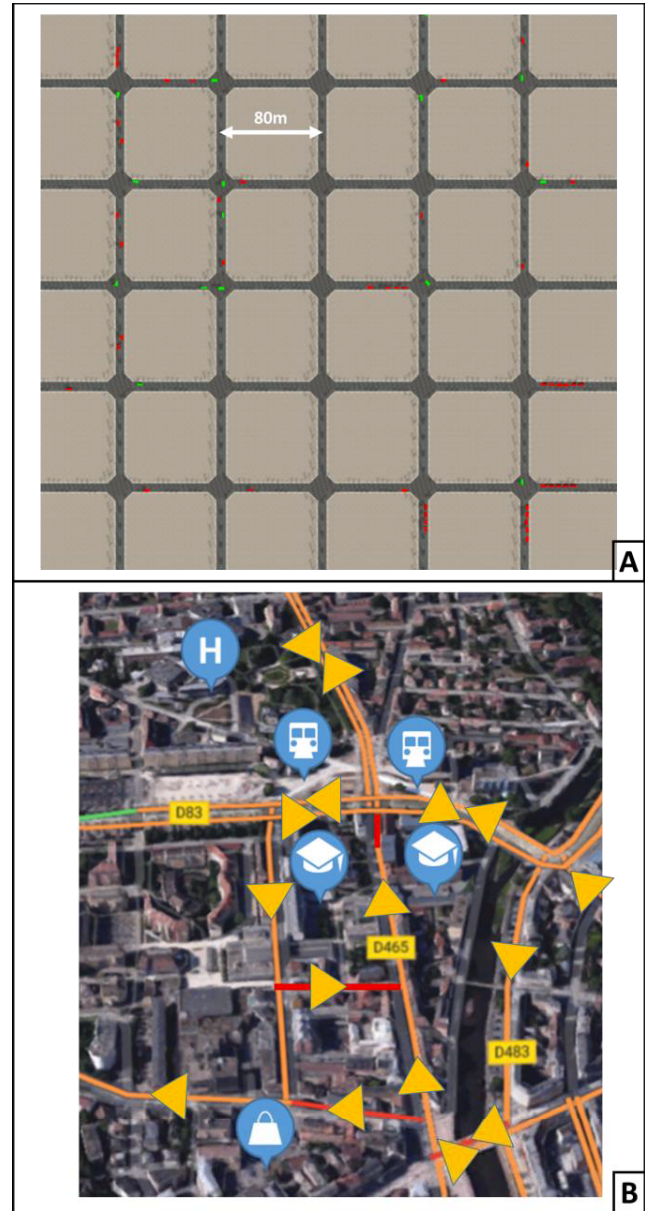


Fig. 17. Simulated networks. A- Grid Network, 25 linked intersections, each lane is unidirectional, two directions are available between each pair of intersections and 10 available spaces per lane between two intersections. B- Urban network, Place Yitzhak Rabin, (90000 Belfort) is crossed by 6000 vehicles and served by more than 39 buses at rush hours.

considering bidirectional lanes as shown in Fig.10. This makes the path update capable of changing the direction of roads according to the traffic, which is a definite advantage that other traffic management system do not possess. Hence, in the following, only the heuristic without any improvement is considered.

V. SIMULATIONS AND RESULTS

Simulations aim at evaluating our approach in terms of traffic deadlock prevention, traffic efficiency, and computation performance. We used a traffic simulator developed by Voxelia (Fig.17-A) for simulating connected and autonomous vehicles. Virtual vehicles are submitted to physical constraints and use

the well-known Intelligent Driver Model (IDM) [39]. The IDM is improved in order to respect a bounded acceleration between -2 m/s^2 and 2 m/s^2 . As a result, the simulated autonomous vehicles have a smooth driving style.

In order to accurately evaluate the performance of the presented algorithm, we compare our approach to the ordinary traffic controlled by traffic lights. We use VISSIM to simulate human drivers. VISSIM has been widely accepted simulation tool for traffic engineering purposes due to its realistic representation of traffic of human-driven-vehicles under a variety of scenarios [41].

There are two studied networks. The first one is a grid composed of 25 interconnected intersections. The length between each pair of neighbor intersections is set to 80 meters (Fig.17-A). The other one, is a real urban network served by buses and private vehicles (Fig.17-B). This network links the residential area to two schools, offices, a hospital and a shopping area. The first network is called grid network whereas the second one is called urban network.

A. Network Management

In all performed simulations of autonomous vehicles, we consider that the Distributed Clearing Policy (DCP) is used at each intersection, for deciding which vehicle has the right-of-way [12]. In very simple words, DCP operates according to the two following rules:

- If there is no collision risk, DCP evacuates the vehicle as soon as possible,
- DCP avoids cutting a convoy of vehicles.

DCP is used because it provides a high intersection throughput while being resistant to message loss.

The following four protocols are compared: Cooperative Vehicle-Actuator System (CVAS) [14], Kanban, CVAS for Network of Intersections (CVAS-NI) and traffic lights. DCP is used in the first three protocols for which we assume a traffic of connected and autonomous vehicles.

1) *CVAS*: The control is independently performed at each intersection. In other words, there is no server to manage the entire network. Intersection servers do not communicate together. If both DCP rules are respected, the vehicle obtains the right-of-way.

2) *Kanban*: It respects the occupancy constraints through tokens. Each lane that connects two intersections owns a maximal number of tokens which represents the number of free spaces. Neighbor intersections communicate together to control the occupancy of lanes. A right-of-way is given only if a token is available for the desired destination. As a result, Kanban does not allow vehicles to cross the intersection if the next lane is full.

We call it Kanban because it is inspired from the well-known order system used in manufacturing. Kanbans are tokens used to signal from the downstream to the upstream station that parts are needed. This ensure that parts would only be produced if they would actually be used at a later stage of production (see [40]).

3) *CVAS-NI*: It is entirely based on the protocol described in this paper. The network server manages all the occupancy and reservation constraints.

The main objective of comparing CVAS-NI to CVAS and to Kanban is to assess the deadlock prevention efficiency. Neither CVAS nor Kanban are able to prevent deadlock. However, when all vehicles go straight, Kanban coupled with DCP produces green waves. Indeed, empty lanes call for the traffic and so on, until vehicles reach the exit zone.

4) *Traffic Lights*: In VISSIM, the grid is managed by traffic lights. The cycle duration is empirically optimized to maximize the throughput, by delaying deadlock situation to the maximum amount of time. The traffic lights are synchronized to maximize the bandwidth. For the urban network, the traffic lights have the same cycle times, splits and offsets as reality. The movements at the conflict zones are managed through different levels of priorities to avoid collisions. More details about VISSIM management of conflict zones are provided in [33].

B. Generated Sequence

For each test, we randomly generate a sequence of a given number of vehicles. Each vehicle is generated at a given time for a known path. Thus, for each sequence, we can reliably compare the four protocols mentioned previously. Once the four protocols are tested for the same generated sequence, a new sequence is generated to create a different simulation and so on. We then compute the average values of 10 simulations of the same policy. There are two considered probabilistic events:

- New vehicle in the network: the generation of vehicles in each entrance zone of the network follows an exponential distribution with the parameter λ being the flow rate per lane given in vehicles per lane per second (vls).
- Vehicle direction: at each intersection, a vehicle has probabilities to turn right, left or to go straight. The two first probabilities have the same value. Hence, it is sufficient to mention the probability to go straight.

A high probability of turning increases the risk of having a circular dependency between vehicles. Moreover, vehicles may have cycles in their itineraries. Also, turning movements generate more conflicts. Thus high probability of turning significantly increases the length of itineraries and the duration of grid occupancy.

C. Simulation Results: Grid-Network

First, we have compared the four policies considering that all vehicles go straight (Fig.18-A) with $\lambda = 0.2 \text{ vls}$ (vehicles per lane per second) and we generate 5000 vehicles. The presented number of vehicles includes vehicles that are waiting in the entrance zones. One can note that CVAS policy leads to deadlock very quickly. Kanban, CVAS-NI and traffic lights (VISSIM) behave similarly. The three create green waves for evacuating vehicles as soon as possible. CVAS-NI, as well as Kanban have fewer vehicles in the grid because they are more

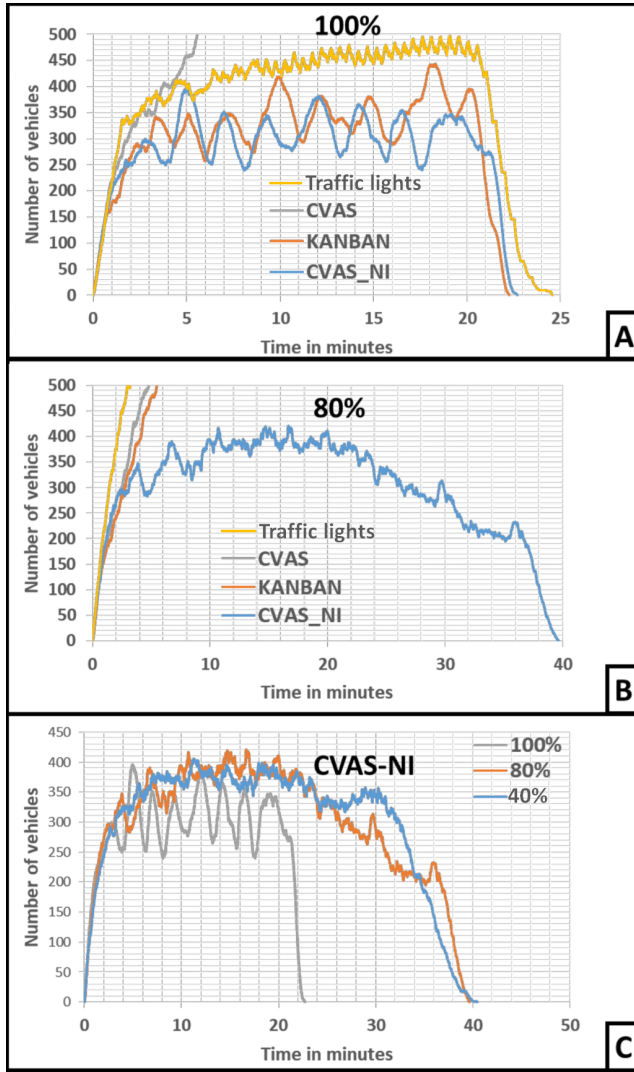


Fig. 18. Simulation results, A- 4 Intersection managements compared through the number of vehicles in the network when all vehicles go straight. B- 4 Intersection managements compared through the number of vehicles in the network when 80% of vehicles go straight. C- The behavior of CVAS-NI according to the probability of going straight.

efficient for evacuating vehicles at intersections. Once the generation is finished, between 2 and 3 minutes are necessary to evacuate the remaining vehicles.

Secondly, we consider that each vehicle has a probability to turn at each intersection either left or right, with $\lambda = 0.2$ vls. Fig.18-B shows that besides CVAS that fails previously, neither Kanban nor traffic lights (VISSIM) are able to manage the network. Fig.18-C shows that for different probabilities of turning left or right there is no deadlock with CVAS-NI. The network throughput is relatively stable during the simulation. The probability of turning provokes an increase of vehicles in the network because the traveled distance is greater. However, in all cases, CVAS-NI succeeds. The throughput and the average speeds are given in Table I.

In the following, we consider less traffic, with $\lambda = 0.05$ vls and the probability to go straight equals 80%. The main objective of the simulations is to assess the behavior of the CVAS-NI when there is no risk of deadlock. The results are

TABLE I
THE EFFECT OF THE PROBABILITY OF TURNING
MOVEMENT ON TRAFFIC PERFORMANCE

	100%	80%	40%
Throughput	0.2 vls	0.13 vls	0.11 vls
Average speed	29.98 km/h	15.33 km/h	13.42 km/h

TABLE II
GRID NETWORK: COMPARISON BETWEEN THE FOUR TRAFFIC
MANAGEMENT APPROACHES AT LOW TRAFFIC

Traffic management	Performance	Results
CVAS	Average number of vehicles	51.8 vehicles
	Average speed	42.17 km/h
	Average stop time	1.03 s
Kanban	Average number of vehicles	42.3 vehicles
	Average speed	42.12 km/h
	Average stop time	1.01 s
CVAS-NI	Average number of vehicles	48.2 vehicles
	Average speed	36.78 km/h
	Average stop time	12.5 s
Traffic lights	Average number of vehicles	97.5 vehicles
	Average speed	29.98 km/h
	Average stop time	38.57 s

TABLE III
SIMULATION RESULTS OF PLACE YITZHAK RABIN AT RUSH HOURS

Traffic management	Performance	Results
CVAS	Average number of vehicles before deadlock	247 vehicles
	Average throughput before deadlock	0.09 vls
	Average time without deadlock	15.5 minutes
	Stuck vehicles	289 vehicles
	Stuck buses	28 buses
Kanban	Average number of vehicles before deadlock	234 vehicles
	Average throughput before deadlock	0.11 vls
	Average time without deadlock	25 minutes
	Stuck vehicles	143 vehicles
	Stuck buses	8 buses
CVAS-NI	Average number of vehicles	158 vehicles
	Average throughput	0.09 vls
	Average time without deadlock	no deadlock
	Stuck vehicles	no deadlock
	Stuck buses	no deadlock
Traffic lights	Average number of vehicles before deadlock	611 vehicles
	Average throughput before deadlock	0.10 vls
	Average time without deadlock	15.2 minutes
	Stuck vehicles	310 vehicles
	Stuck buses	24 buses

presented in Table II. One can observe that CVAS-NI performs better than traffic lights, but worse than CVAS and Kanban. This is due to the fact that there is no optimization as discussed in subsection IV.E.1. The matrix is filled according to the vehicle arrivals and once a vehicle is in the matrix there is no updates of its path.

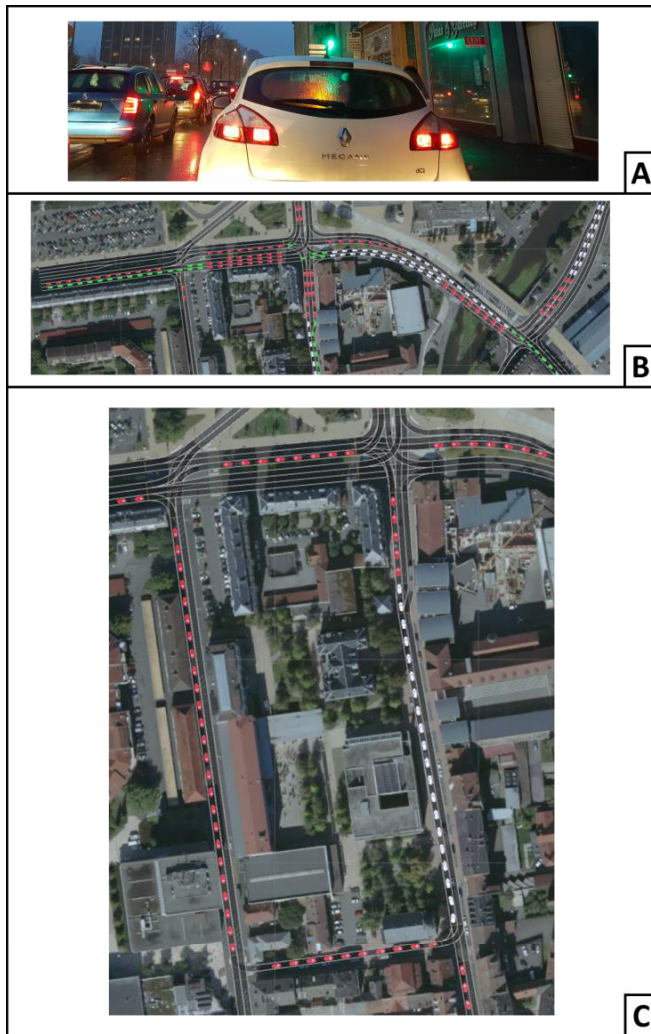


Fig. 19. Deadlock in Place Yitzhak Rabin, A- Real deadlock, B- CVAS deadlock due to occupied box junctions (Simulation). C- Kanban deadlock due to filled circuit (Simulation).

D. Simulation Results Urban-Network

Place Yitzhak Rabin in Belfort is a good example where deadlock can take place at rush hours (see Fig.19-A). There are two patterns of deadlock similar to the ones presented in Fig.1-C and Fig.2-A. The human drivers mainly yield the right to pass or tighten the queue in order to slowly recover movement.

The results of simulations are presented in Table III. The average number of vehicles as well as throughput are given before deadlock happens for CVAS, Kanban and traffic lights. Deadlock systematically happens in both CVAS and traffic light scenarios. In CVAS deadlock occurs because of the filled box junctions (see Fig.19-B). In the Kanban scenario, deadlock happens variably, on average after 25 minutes. The deadlock is due to filled circuit to drop off children at school (see Fig.19-C). In VISSIM simulations, we have both deadlock situations, i.e. filled box junctions after 15.2 minutes of simulation and the filled circuit without possible movement after 30 minutes of simulation. The latter deadlock happens since VISSIM removes a few stuck vehicles, which allows others to fill the circuit to its blockage.

One can observe that CVAS-NI is slightly less efficient than traffic lights and Kanban before deadlock. This is due to both facts: paths of vehicles in the matrix are not updated and there are many turn movements. These same facts are the reason why CVAS and Kanban perform better in the empty grid. However, even if paths are not updated, the simulation shows that the performance of CVAS-NI is still close to the other traffic management policies, while keeping a deadlock-free network.

E. Computation Performance

We use a laptop Intel® Core™ i7-3537U CPU 2.5 Ghz. In all simulations, vehicles are added to the matrix in less than 1 ms. In the Grid-Network, if the arrival rate is 0.5 vls we had the worst cumulative cpu-time of 20 ms for adding several vehicles simultaneously.

VI. CONCLUSION

We have proposed a protocol able to efficiently prevent the deadlock in a network of intersections. The protocol is simulated in an environment with a high risk of deadlock. The main principle is the reservation of each vehicle journey through the network according to space and time. The method is only partially centralized. We proposed a hierarchical approach where network servers, intersection managers and vehicles all together contribute to the decision making process of improving the traffic efficiency. Moreover, the duty of each collaborative entity is limited to a few rules, in order to respect real-time constraints. The reservation is applied only on the roads that connect intersections or grids together, thus each intersection manages the traffic locally to improve the overall throughput without deadlock. As a result, the computation time required is low, while network capacity is increased. In this paper the server adds vehicles to the matrix without improving their paths and schedules. The algorithm offers various ways to reconsider the vehicle paths, which deserve to be investigated.

VII. PERSPECTIVES

Our work aims to prepare the widespread adoption of autonomous vehicles, in the near future. The autonomous vehicle should progressively become omnipresent in cities. Some cities will limit the access of dense area to autonomous vehicles only (Eg. Masdar project). Therefore it is important to prevent deadlock.

The protocol can be used as a base for a few main perspectives:

1) *Microscopic Simulations*: Many traffic scenarios cannot be studied in depth because traffic simulators suffer from deadlock situations. Indeed, simulators can quickly experience gridlock, which prevents the simulation from terminating. The proposed protocol offers the possibility to run simulations under any traffic configuration as no deadlock can appear. Hence, it becomes possible to study new approaches for controlling the traffic of connected and autonomous vehicles. For instance, we can consider different itinerary optimization algorithms [42], in order to improve the efficiency of the whole network [38]. These strategies can be used as a new way for

controlling the traffic in the city by rerouting vehicles. Besides, the protocol can be used to study the grid configuration [43], road reservation [44], [45], dynamic lane reversal [46] and even allows to extend the concepts through a real-time negotiation of the infrastructure utilization. For this purpose, we draw the reader's attention to the fact that the reserved path is one possible solution. Thus, one vehicle is able to modify a part of its itinerary if it is allowed by the server.

2) *Parking Lots*: Areas to park vehicles are at the core of the space sharing problem in cities. Indeed, one vehicle can be stationed for a long undetermined period of time. If one or more parking lots are full, then it provokes the congestion formation as drivers have to search free places without guarantee of finding them. It is then interesting to include parking lots in the protocol to study strategies of parking management.

3) *AGV and PRT*: The increasing number of automated guided vehicles (AGV) in the industry requires policies to avoid collisions and deadlocks.

Usually, these systems prevent them by avoiding intersections of assembly lines by design. Sometimes, intersections are required, and a traffic control solution limits the presence of AGV via a zone-control policy. Both solutions limit the number of AGV in use.

With cooperative intersection management, we allow new designs for lines of AGV [19], [47], [48]. The intersections are not a problem anymore as the regulation policy prevents collisions, deadlocks, and gridlocks. Therefore, the number of operating AGV can be increased at almost no cost (given that they usually are already equipped with communication units).

Furthermore, we can apply the same principle to personal rapid transit (PRT) to prevent deadlock, hence allowing to have a fully automated system of vehicles without the need of human regulation.

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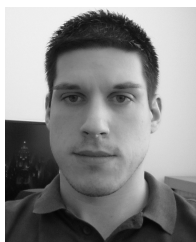
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Florent Perronnet received the Ph.D. degree in automatic control from Université de Technologie de Belfort-Montbéliard, France, in 2015. He is currently a Researcher with LE2I-UBFC-UTBM and he is working on the project Explorys in partnership with Alstom. His research interests include multi-agent simulations, discrete event systems, wireless systems applied to the traffic control, and the possible use of autonomous vehicles. His research deals with the problem of urban congestion, in particular at intersections shared with public transport vehicles.

The studied solutions use new technologies in communication systems and positioning systems in order to develop cooperative intersection management.



Jocelyn Buisson received the Ph.D. degree in computer science from Université de Technologie de Belfort-Montbéliard, France, in 2014. He is currently a Researcher with LE2I-UBFC-UTBM and he is working on the European project SAFER-LC. His research interests include multi-agent-based simulations of traffic and pedestrian in urban environment and autonomous vehicles.



Alexandre Lombard received the Ph.D. degree in computer science from the Université de Bourgogne Franche-Comté, France, in 2017. He is currently Associate Professor with LE2I-UBFC-UTBM. He was a Researcher on the project Astres, in partnership with Alstom, for two years. His research interests include intelligent transportation systems, cooperative traffic management, and autonomous vehicles. His research is mainly focused on the usage of wireless vehicular communication to improve the traffic efficiency, in particular at intersections. He participated in the X.icars demonstration at ITS WC 2015 where 3 driverless vehicles synchronized their movement into an eight-shaped circuit.



Abdeljalil Abbas-Turki received the Ph.D. degree in control and computer science from the Université de Technologie de Belfort-Montbéliard (UTBM), France, in 2003. He is currently an Associate Professor at LE2I-UBFC-UTBM. His research interests include discrete event dynamic system and combinatorial optimization applied to the urban traffic modeling and control. He is involved in many bus rapid transit projects, for studying the future behavior of the traffic through macroscopic and microscopic models. He is also involved in developing cooperative intersection management for encouraging new mobility systems and in timetabling urban rail traffic. He was the Head of the X.icars demonstration at ITS WC 2015, held in Bordeaux, France.



Mourad Ahmane received the Ph.D. degree in automatic control and computer science from the Ecole Centrale de Nantes, France, in 2006, after obtaining the master's degree in 2003. Since 2009, he has been a Researcher with the Université de Technologie de Belfort-Montbéliard. His research interests include intelligent transportation systems, manufacturing systems, discrete event dynamic systems, dioid algebra, and stochastic systems. He is involved in the development of cooperative intersection management with autonomous vehicles in order to enhance new mobility systems.



Abdellah El Moudni received the Ph.D. degree in automatic control from Lille University, Lille, France, in 1985. He is currently a Professor with the Université de Technologie de Belfort-Montbéliard, Belfort, France, where he is in charge of courses in automatic control. His research interests include transportation systems, nonlinear control theory, dynamic systems, control of discrete event systems, and singular perturbation methods in discrete time.